Attention-Based Safe Reinforcement Learning for Scale-Agnostic and Constraint-Guaranteed V2G Charging Control

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Abstract

Charge point operators that offer vehicle-to-grid (V2G) services must schedule many electric vehicles (EVs) under a transformer limit and deliver the energy that each driver requests by departure. Reinforcement learning (RL) controllers for this task usually treat both requirements as reward penalties, and their policy networks are tied to one station size. This paper proposes a safe attention policy (SA-Attn) that joins a size-agnostic policy with a closed-form safety layer. Each charger is a token of a transformer encoder, and a shared head maps every token to a raw power action, so one set of weights serves any number of chargers. The safety layer projects the raw actions onto a feasible set defined by per-EV power bounds and by the forward and reverse transformer limits. Its lower bound for each EV is the least power that still lets full-power charging reach the departure energy, computed from a reachable-energy operator that includes the charging taper. The projection reduces to clipping each action around one common shift, found in $O(N \log N)$ time for N chargers.

Under stated assumptions, the paper proves that the transformer limit and every departure energy requirement hold at every step for any raw action, during training and deployment. A least-laxity-first fallback rule and admission control handle steps on which the assumptions fail, and these steps are counted. Uncertain departure times are handled by a split-conformal lower bound that turns the guarantee into a coverage-level statement. The policy is trained with proximal policy optimization (PPO) on stations with 10 and 20 chargers in the EV2Gym simulator. Without retraining, it is evaluated on stations with up to 200 chargers, three arrival profiles and two transformer ratings. Performance is reported as the optimality gap to a perfect-information linear program with hard constraints, and the cost of the guarantee as the profit difference to a penalty-trained attention policy. [RESULT: one or two sentences with the main findings: optimality gap at $N = 200$, constraint satisfaction and infeasible-step frequency, cost of the guarantee]

Keywords: Attention mechanism, Electric vehicle charging, Safe reinforcement learning, Vehicle-to-grid, Zero-shot transfer

1 Introduction

A charge point operator (CPO) buys energy for its chargers at time-varying prices, and vehicle-to-grid (V2G) discharging lets the same operator sell energy from parked electric vehicles (EVs) when prices are high. Two hard requirements limit this freedom. The aggregate power of the station must stay within the transformer limit, and each EV must hold its requested energy when it leaves. A schedule that breaks the first requirement overloads equipment, and one that breaks the second requirement breaks the service promise to the driver.

Reinforcement learning (RL) has become a common approach to this control task, because prices, arrivals and departures are uncertain and hard to model in advance [1, 2]. Most RL controllers for EV charging encode the departure energy requirement and the transformer limit as weighted penalties in the reward [3–6]. A penalty trades violations against profit, so a trained policy may still violate either requirement at test time [6, 7]. Constrained formulations replace the fixed weight with a bound on the expected cumulative cost [8], but such a bound holds on average over episodes and not at every step [9, 10].

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A second limitation concerns station size. The default interface of the EV2Gym simulator assigns one action to each charger and uses a flat observation, so a trained network is tied to one number of chargers [6]. Graph and attention policies remove this restriction [11–13], yet in these size-agnostic controllers the constraints still enter through reward terms, imitation data or heuristic filters. The authors of one of them state explicitly that their filter makes no pointwise feasibility claim [13].

Safety layers correct each action before execution and can enforce limits by construction [7, 14, 15]. For EV stations, existing layers cover per-step limits such as transformer capacity and charger current [16], or they apply rule-based shields [17, 18]. A per-step layer alone does not protect the departure energy requirement, because that requirement couples the present action with every later step of the stay. Per-EV latest-start bounds address this coupling in laxity-based scheduling [19] and in hierarchical V2G scheduling [20]. To our knowledge, no published controller combines such bounds with a learned policy that runs at any station size and comes with a proof of multi-step feasibility. [TODO: Confirm this sentence against the full texts of [21], [22] and [23] before submission (novelty risks listed in the gap analysis)].

This paper proposes a safe attention policy (SA-Attn) for V2G charging control at a CPO station. An attention policy treats each charger as a token and outputs one raw power action per token with shared weights. A safety layer, a closed-form projection onto the feasible set, maps the raw actions to commanded powers that respect the per-EV bounds and the transformer limit. The projection has the same form as the rate rule of smoothed least-laxity-first (sLLF) scheduling [19], and it is the classical Euclidean projection onto a box intersected with one sum constraint [24, 25]. Our contribution therefore lies in the bounds fed to this projection, in its use during training and deployment, and in the multi-step argument built on it.

The paper makes four contributions, and each of them corresponds to a part of the evaluation in Section 6.

1. We design an attention policy for V2G charging control in which each charger is a token and a shared head produces one action per token. The policy is permutation-equivariant, ignores empty chargers by masking and has a parameter count that does not depend on the number of chargers.
2. We derive a safety layer whose per-EV lower bound follows from a reachable-energy operator that includes the charging taper. Theorem 1 proves that the transformer limit and every departure energy requirement hold at every step for any raw action, under stated assumptions. A least-laxity-first fallback rule and admission control handle the steps on which aggregate feasibility fails.
3. We define a zero-shot evaluation in EV2Gym that trains on 10 and 20 chargers and tests on up to 200 chargers without retraining. Performance is measured as the optimality gap to a perfect-information linear program with hard constraints, and the cost of the guarantee is measured against penalty-based training of the same network.
4. As a secondary contribution, we replace the known-departure assumption with a split-conformal lower bound on the departure step, and Proposition 2 turns the guarantee into a coverage-level statement.

Section 2 shows that none of these ingredients is new in isolation, so the claim of the paper rests on their combination and on the measured price of the guarantee.

Section 2 reviews related work, and Section 3 states the system model and the Markov decision process (MDP). The attention policy, the safety layer and the training algorithm follow in Section 4, and Section 5 describes the experimental setup. Results appear in Section 6 and the discussion in Section 7. Section 8 concludes the paper, and the supplementary material contains all proofs, the EV2Gym modelling details, the training details and further result tables.

2 Related Work

2.1 Reinforcement learning for EV charging and V2G

Deep RL for EV charging has moved from single-vehicle agents [26] to stations with photovoltaics, storage and V2G [5, 18], to coordinated stations [22] and to fleets under distribution-grid limits [27, 28]. Two

reviews list safety, dependence on simulation and the lack of a standard benchmark among the open problems [1,2]. In most studies, the energy owed to a departing vehicle enters the reward as a penalty [3–5].

EV2Gym models a CPO whose chargers sit under transformers with inflexible load, solar generation and demand response [6]. Its default RL interface uses one action per charger port and a flat observation, so a trained policy is tied to one station size. The simulator records transformer overload without preventing it, and in its V2G benchmark penalty-based agents earned profit but left many users short of their requested energy. Optimisation-based controllers treat both requirements as hard constraints, and the Adaptive Charging Network (ACN) solves a convex program in receding horizon [29]. Economic model predictive control (MPC) with hard departure and transformer constraints is also available inside EV2Gym [30]. Recent EV2Gym-based work trains actors through a differentiable power-flow model, but voltage and departure targets remain soft penalties there [31].

2.2 Constraint handling and safe reinforcement learning

Weighted violation terms in the reward do not prevent violations at test time [7, 32]. The constrained Markov decision process (CMDP) replaces the fixed weight with a bound on expected cumulative cost [8], solved by Lagrangian [33] or trust-region [34] updates. Reviews classify both approaches as soft or probabilistic guarantees [9, 10], and CMDP formulations have been applied to EV charging with voltage limits [32], transformer thermal stress [35] and V2G aggregators [23].

A second family corrects each action before execution, through shields [36], learned linear safety layers [14] or quadratic programs with a fallback controller [37]. EV examples include rule-based shields [17], action-replacement layers [18] and grid-code shielding [38]. Such a filter is only as safe as its internal model: HetGPS combines a learned risk model with a linearised power-flow projection, yet its authors make no pointwise feasibility claim [13]. The EV2Gym authors have also added a safety layer to a graph-enhanced twin delayed deep deterministic (TD3) agent [39]. According to its abstract, the layer preserves charging requirements and enforces power limits, but the abstract describes neither the mechanism nor a formal guarantee [21]. [TODO: Describe the layer of [21] precisely once the full text is read].

A third family makes the policy output feasible by construction. Differentiable convex optimisation layers pass gradients through an exact solve [40, 41], and Chen et al. used such a projection inside energy-system policies [15]. Gauge maps give feasible points in closed form [42], and Tabas and Zhang used one to prove that a learned frequency controller stays in a controlled-invariant safe set [7]. For EV stations, Wan et al. prove per-step feasibility of a quadratic-program projection that covers transformer and charger-current limits but not departure energy [16]. Yan et al. derive per-EV bounds from remaining energy and time, and in the version we read their formal result covers only first-order power balance [20]. A depot-charging preprint reports no breaches with a closed-form projection but states no proof in its abstract [43].

Projection also alters learning, because naive projection biases policy gradients [44] and many infeasible actions map to the same projected point [45]. For stochastic policies, projection inside the environment is equivalent to projection as a policy layer [46], and a distance penalty between pre- and post-projection actions limits drift of the raw actions [15, 46].

2.3 Size-agnostic architectures

Deep Sets [47], self-attention [48] and the Set Transformer [49] process sets of any size with shared weights, and UPDeT decoded entity-wise actions so that one model served multi-agent tasks of different sizes [50]. Size transfer is not automatic, since neural routing solvers trained on small graphs often fail on larger ones [51]. In EV charging, aggregation gave the first size-independent states. Sadeghianpourhamami et al. binned vehicles by time to departure and remaining charging time, reported the gap to a perfect-knowledge optimum and tested scaled arrival rates [3]. Aggregated representations, however, hide per-vehicle detail or move it into disaggregation rules [52, 53].

Parameter sharing keeps the per-vehicle detail. The heterogeneous graph policy (EV-GNN) of [11] scales to 1000 charge points with a reward that penalises transformer overload and unmet demand. In contrast, the graph-enhanced decision transformer (GNN-DT) of [12] transfers zero-shot from 25 to 100 chargers, with performance degrading as size grows. HetGPS keeps one parameter set from 1,236 to 3,218 vehicles [13], and a graph actor for voltage control has been deployed on a larger feeder without retraining [54]. In

Table 1: Positioning against the closest works. "Penalty" means a reward term; "hard" means enforced by construction. Entries rest on full-text readings except where marked (abstract only).

Work	Size-agnostic policy	Departure energy	Transformer limit	Guarantee	Zero-shot size test	Gap to optimum
EV2Gym agents [6]	no (flat)	penalty	penalty	none	no	oracle as baseline
EV-GNN [11]	yes (graph)	penalty	penalty	none	not in version read	oracle or MPC baseline
GNN-DT [12]	yes (graph)	penalty	penalty	none	25 to 100 chargers	reward within 5% of oracle
HetGPS [13]	yes (graph attention)	reward and heuristic	not constrained	empirical (voltage filter)	8 to 32 transformers	no
Yan et al. [20]	no	hard (per-EV rule)	not modelled	first-order power balance	no	gap to MPC
Wan et al. [16]	no	not modelled	hard (quadratic-program projection)	per-step feasibility	no	no
sLLF [19]	rule, any size	laxity rule	hard (common level)	feasibility theorems	not learned	no
Sadeghian-pourhamami et al. [3]	yes (aggregate state)	penalty	not modelled	none	scaled arrivals	yes
Orfanoudakis et al. [21] (abstract only)	graph policy	safety layer	safety layer	unknown	unknown	unknown
This work	yes (charger tokens)	hard (bound l_i)	hard (common shift)	proven, Assumptions A1–A5	10/20 to 200 chargers	hard-constrained linear program; cost of the guarantee

all size-agnostic controllers that we could read in full, constraints enter through reward terms, imitation data or heuristic screens.

2.4 Laxity-based scheduling and departure uncertainty

Chen et al. proved that a schedule is feasible exactly when every laxity, the slack between the time left and the time still needed to finish charging, stays non-negative [19]. Their sLLF rule clips each rate around a common level found by bisection under the station power limit, and their resource-augmentation bounds indicate that online feasibility generally requires extra power. Least-laxity-first (LLF) priorities also perform well under hard cable capacities [55] and in rule-based aggregators that avoid transformer overload [56]. These methods assume that users declare departure time and energy at arrival. Residential plug-in durations are hard to predict [57], users can gain by under-reporting deadlines [58], and forecasts of departure improve prioritisation under scarce supply [59]. Conformal prediction gives distribution-free coverage under exchangeability [60,61] and has tightened constraints in safe planning [62]. Our screening, however, found no peer-reviewed work that feeds a calibrated departure bound into an RL safety layer for EV charging.

2.5 Positioning

As seen in Table 1, each ingredient of our design already exists in some form. Size-agnostic graph and attention policies with zero-shot transfer exist for EV charging [12, 13], and provable projection layers exist in safe RL [7, 16]. The clip-around-a-common-level rule appears in sLLF [19], and per-EV latest-start bounds appear in the scheduler of Yan et al. [20]. Train-small, test-large evaluation with a gap to an optimum dates back at least to the binned-state study of [3]. To our knowledge, no prior work joins these elements in one controller with a proof of multi-step feasibility. SA-Attn uses one set of weights for any number of chargers, and its safety layer keeps departure energy and the transformer limit feasible at every step under stated assumptions, as we prove. Zero-shot results up to 200 chargers are reported as gaps to a hard-constrained perfect-information optimum, together with the cost of the guarantee.

3 Problem Formulation

3.1 Station, time and prices

A CPO runs a station with N chargers $\mathcal{N} = \{1, \dots, N\}$ behind one transformer, and each charger hosts at most one EV. While an EV is connected, its quantities carry the index i of its charger. Time is divided into steps $t = 0, \dots, T-1$ of length Δt [h], and the action of step t is held constant on $[t\Delta t, (t+1)\Delta t)$. At every step the station observes the transformer rating $P_{\mathrm{tr}}(t)$, the inflexible load $P^{\mathrm{inf}}(t) \geq 0$ and the photovoltaic (PV) generation $P^{\mathrm{pv}}(t) \geq 0$, all in kW. Energy is bought at the price $c^{\mathrm{ch}}(t)$ and sold at the price $c^{\mathrm{dis}}(t)$ [€/kWh], and both prices may be negative.

3.2 EV energy dynamics

EV i arrives at step t_i^{arr} and is controlled at the steps $t = t_i^{\mathrm{arr}}, \dots, d_i$, where the departure step d_i is the last controlled step. The EV leaves after the action of step d_i has been applied, so with $E_i(t)$ denoting the battery energy at the start of step t, the departure energy is $E_i(d_i + 1)$. At step t, the number of remaining actions, including the current one, is

$$\tau_i(t) = d_i - t + 1, \tag{1}$$

and the set of occupied chargers is $\mathcal{N}(t) = \{i : t_i^{\mathrm{arr}} \leq t \leq d_i\}$, with $n(t) = |\mathcal{N}(t)|$ connected EVs.

Let $p_i(t)$ [kW] be the grid-side power of charger i, positive for charging and negative for V2G discharging. With $p^+ = \max(p, 0)$ and $p^- = \max(-p, 0)$, the battery energy evolves as

$$E_i(t+1) = E_i(t) + \Delta t\, g_i\big(p_i(t)\big), \qquad g_i(p) = \eta_i^{\mathrm{c}} p^+ - p^- / \eta_i^{\mathrm{d}}, \tag{2}$$

where $\eta_i^{\mathrm{c}}, \eta_i^{\mathrm{d}} \in (0, 1]$ are the charge and discharge efficiencies, and the energy map g_i is continuous, strictly increasing and piecewise linear with $g_i(0) = 0$. Per-step physical limits restrict the power to

$$p_i^{\min}(E) \leq p_i(t) \leq \bar{P}_i(E), \qquad p_i^{\min}(E) = -\min\Big\{P_i^{\mathrm{dis}},\ \eta_i^{\mathrm{d}} \frac{(E - E_i^{\min})^+}{\Delta t}\Big\}, \qquad E = E_i(t), \tag{3}$$

where $\bar{P}_i(E) \geq 0$ is the state-dependent maximum charging power. This maximum includes the constant-current, constant-voltage taper and the capacity limit $\eta_i^{\mathrm{c}} \bar{P}_i(E) \Delta t \leq E_i^{\mathrm{cap}} - E$, and its nominal value is $\bar{P}_i^{\mathrm{nom}} = \sup_E \bar{P}_i(E)$. The maximum discharging power is $P_i^{\mathrm{dis}} \geq 0$, and under Equation 3 the energy stays in $[E_i^{\min}, E_i^{\mathrm{cap}}]$.

The theory of Section 4 holds for the general map g_i, but EV2Gym reports, bills and meters the battery-side energy change as the charger power [6]. Our experiments therefore use $\eta_i^{\mathrm{c}} = \eta_i^{\mathrm{d}} = 1$, so that $g_i(p) = p$ and EV2Gym's own efficiencies act only as derating factors on the power limits. This convention removes the round-trip loss from V2G arbitrage, and Section 7.3 lists it as a limitation.

3.3 Departure energy requirement

Each EV requests a desired departure energy E_i^{des}, and at arrival the request is capped to the energy that is individually reachable during the stay,

$$E_i^{\mathrm{req}} = \min\Big\{E_i^{\mathrm{des}},\ E_i^{\mathrm{arr}} + R_i\big(\tau_i(t_i^{\mathrm{arr}}), E_i^{\mathrm{arr}}\big)\Big\}, \tag{4}$$

where $E_i^{\mathrm{arr}} = E_i(t_i^{\mathrm{arr}})$ and $R_i(k, E)$ is the reachable-energy operator of Section 4.2. Without this cap, some arrivals could not be served by any controller, and a guarantee about them would be void. The departure energy requirement of EV i is therefore

$$E_i(d_i + 1) \ \geq \ E_i^{\mathrm{req}}. \tag{5}$$

In code, the cap uses the same certified threshold routine as the safety layer and lowers the value of Equation 4 by at most 10^{-6} kWh. Admission control (Section 4.5) may lower the requirement of a newly arriving EV further, and the experiments report the number and size of capped and reduced requests.

3.4 Transformer limit

The net transformer power $P^{\mathrm{inf}}(t) - P^{\mathrm{pv}}(t) + \sum_i p_i(t)$ must stay within $[-P_{\mathrm{tr}}(t), P_{\mathrm{tr}}(t)]$, and for the EV aggregate this symmetric thermal limit gives the transformer limit

$$-C_{\mathrm{rev}}(t) \le \sum_{i \in \mathcal{N}} p_i(t) \le C(t), \qquad C(t) = P_{\mathrm{tr}}(t) - P^{\mathrm{inf}}(t) + P^{\mathrm{pv}}(t), \qquad C_{\mathrm{rev}}(t) = P_{\mathrm{tr}}(t) + P^{\mathrm{inf}}(t) - P^{\mathrm{pv}}(t). \tag{6}$$

The forward limit $C(t)$ bounds charging and the reverse limit $C_{\mathrm{rev}}(t)$ bounds V2G export, and their sum $2P_{\mathrm{tr}}(t)$ is never negative. Empty chargers draw zero power, and the safety layer uses the measured inflexible load and PV of the current step. The policy, in contrast, observes only noisy forecasts of future load and PV.

3.5 Online control problem

At each step the CPO chooses $p(t) \in \mathbb{R}^N$ using only the information available at step t and no knowledge of future arrivals. This information comprises the arrival data and current energies of the connected EVs, current and short-window day-ahead prices, and the measured load and PV with their forecasts. The objective is the expected episode profit

$$\max \ \mathbb{E}\Big[\sum_{t=0}^{T-1} \Delta t \sum_{i \in \mathcal{N}(t)} \big(c^{\mathrm{dis}}(t)\, p_i^-(t) - c^{\mathrm{ch}}(t)\, p_i^+(t)\big)\Big], \tag{7}$$

subject to Equations 2–3 at every step, Equation 5 for every EV and Equation 6 at every step.

3.6 Perfect-information optimum

An oracle that knows the whole episode provides the reference for the optimality gap. It sees all sessions with $(t_i^{\mathrm{arr}}, d_i, E_i^{\mathrm{arr}}, E_i^{\mathrm{req}})$, all prices and the realised limits $C(t)$ and $C_{\mathrm{rev}}(t)$. For every session and every step of its stay, the oracle uses charging and discharging powers $p_i^{\pm}(t) \ge 0$, energies $E_i(t)$ and a shortfall slack $z_i \ge 0$ under the constraints

$$\begin{aligned} &E_i(t_i^{\mathrm{arr}}) = E_i^{\mathrm{arr}}, \qquad E_i(t+1) = E_i(t) + \Delta t\big(\eta_i^{\mathrm{c}} p_i^+(t) - p_i^-(t)/\eta_i^{\mathrm{d}}\big), \\ &E_i^{\min} \le E_i(t) \le E_i^{\mathrm{cap}}, \qquad p_i^+(t) \le \bar{P}_i^{\mathrm{nom}}, \qquad p_i^-(t) \le P_i^{\mathrm{dis}}, \qquad E_i(d_i+1) + z_i \ge E_i^{\mathrm{req}}, \\ &-C_{\mathrm{rev}}(t) \le \textstyle\sum_i \big(p_i^+(t) - p_i^-(t)\big) \le C(t). \end{aligned} \tag{8}$$

The linear program (LP) is solved lexicographically with the HiGHS solver [63]: stage 1 minimises the total shortfall $\sum_i z_i$, and its optimum z^* is zero whenever all requirements are jointly feasible. Stage 2 then maximises the profit of Equation 7 subject to Equation 8 and $\sum_i z_i \le z^* + 10^{-6}$, and its optimum is J^{LP}.

The LP relaxes the state-dependent taper to the nominal power $\bar{P}_i^{\mathrm{nom}}$ and allows simultaneous charging and discharging, which never pays when $c^{\mathrm{dis}} \le c^{\mathrm{ch}}$ and is therefore exact in our scenarios. In the exact model, J^{LP} bounds the profit J^π of every controller π that respects Equation 6 and leaves a total shortfall of at most z^*. EV2Gym, however, rounds the stored energy of an active EV up to 0.01 kWh after every step and does not bill this energy. A feasible controller inside the simulator can therefore meet its targets while buying slightly less energy than the exact model requires, and its profit may exceed J^{LP}. We therefore use the rounding-aware optimum $J^{\mathrm{LP,ceil}}$ of the same LP with E_i^{req} and $E_i^{\min}$ lowered by the largest possible rounding gain of 0.01 kWh per step of the stay. Proposition S1 in the supplementary material shows that $J^{\mathrm{LP,ceil}}$ bounds the profit of every such controller inside EV2Gym, and the optimality gap is defined as

$$\mathrm{gap}(\pi) = \frac{J^{\mathrm{LP,ceil}} - J^\pi}{|J^{\mathrm{LP,ceil}}|}. \tag{9}$$

Profits are usually negative, because charging to the requested energy costs money, so the absolute value is needed in the denominator. Episodes with $|J^{\mathrm{LP,ceil}}| < 1\,€$ are excluded from this relative metric, and the gap with respect to J^{LP} is reported in the supplementary material. Because of the relaxations, the reported gap overestimates the true gap; controllers that violate the transformer limit are not covered by the bound and are flagged in the results.

3.7 Markov decision process

We cast the online problem as an MDP with one decision per step, although the problem is strictly partially observable because future arrivals, prices beyond a short window and future loads are not observed. Following standard practice, the policy conditions on the observation described in Section 4.1.

Action. The policy outputs one raw action $a_i(t) \in \mathbb{R}$ per charger, and the action is scaled to kW as $x_i(t) = P^{\text{ref}} a_i(t)$ with $P^{\text{ref}} = 11$ kW. Its executed counterpart $p(t) = \Pi_t(x(t))$ is the output of the safety layer of Section 4.3, or of the fallback rule of Section 4.5 when the feasible set is empty. No squashing function is used, so all bounding is done by Π_t.

Reward. The reward is the realised profit of the step,

$$r(t) = \Delta t \sum_{i \in \mathcal{N}(t)} \left(c^{\text{dis}}(t)\, \tilde{p}_i^{-}(t) - c^{\text{ch}}(t)\, \tilde{p}_i^{+}(t) \right), \tag{10}$$

where $\tilde{p}_i$ is the power reported by the simulator. The reward contains no penalty for the departure energy requirement or the transformer limit, because the safety layer enforces both and such a penalty would vanish on every step where aggregate feasibility holds. On infeasible steps the fallback rule is deterministic, so a penalty there would add variance without a learnable signal. For training, the reward is divided by $N r^{\text{ref}}$ with $r^{\text{ref}} = c^{\text{ref}} P^{\text{ref}} \Delta t$ and $c^{\text{ref}} = 0.25\,€/\text{kWh}$, so that returns do not grow with N and one critic can serve stations of different sizes. All reported results, in contrast, use the undivided profit $r(t)$ in €.

Transition and horizon. Arrivals, prices, loads and PV evolve independently of the actions, and energies follow Equation 2. An arriving EV adds a token and an EV with $d_i = t$ leaves after step t, and since every EV departs before the episode ends, step T is a true terminal state. The discount factor $\gamma = 0.99$ gives an effective horizon of about one day at $\Delta t = 15$ min.

4 Proposed Method

As seen in Figure 1, SA-Attn has three parts that act in sequence at every step. An attention policy maps a set of charger tokens to one raw action per charger, and a safety layer turns the raw actions into commanded powers that respect the per-EV bounds and the transformer limit. During training, a policy-gradient method updates the policy from the realised profit with the safety layer treated as part of the environment, and the same safety layer acts unchanged during deployment.

4.1 Token representation and attention policy

The observation is a set of N charger tokens $\xi_i(t) \in \mathbb{R}^{F_c}$ and one global token $\xi_0(t) \in \mathbb{R}^{F_g}$ with $F_c = 12$ and $F_g = 38$, and no feature encodes the index or the position of a charger. A charger token holds the occupancy flag o_i, the current energy, the capacity, the state of charge, the remaining required energy $(E_i^{\text{req}} - E_i(t))^+$ and the remaining steps $\tau_i(t)$. It also holds the laxity $\chi_i(t)$ of Section 4.2, the nominal charging and discharging powers, the physical lower limit $p_i^{\min}$ and the bounds l_i and u_i of the safety layer. Energies are divided by $E^{\text{ref}} = 100$ kWh, powers by P^{ref} and step counts by the number of steps per day, and an empty charger has an all-zero token.

The global token carries the current prices and the day-ahead charging prices of the next $H_f = 8$ steps, divided by c^{ref}, together with time-of-day and weekend features. It also holds the remaining episode fraction, the current limits $C(t)$ and $C_{\text{rev}}(t)$, the measured load and PV, and noisy forecasts of load and PV for H_f steps. Its last features are the occupancy ratio $n(t)/N$, the rating per charger and the sums $\sum_i l_i(t)$ and $\sum_i u_i(t)$. All grid quantities are divided by the nominal rating $P_{\text{tr}}^{\text{nom}} = \max_t P_{\text{tr}}(t)$, so every global feature is a ratio that does not grow with N. The arrival profile of the station is deliberately left out of the observation, so that the policy can serve stations whose profile is mixed or unknown.

Each token is embedded linearly and receives one of two learned type embeddings, one for charger tokens and one for the global token. The sequence $[e_0; e_1; \ldots; e_N]$ then passes through $n_L = 3$ transformer encoder layers [48] in the pre-LayerNorm arrangement [64, 65], with width $d_m = 128$, $n_H = 4$ heads and no positional encoding. A key-padding mask removes empty chargers as attention keys, but the global

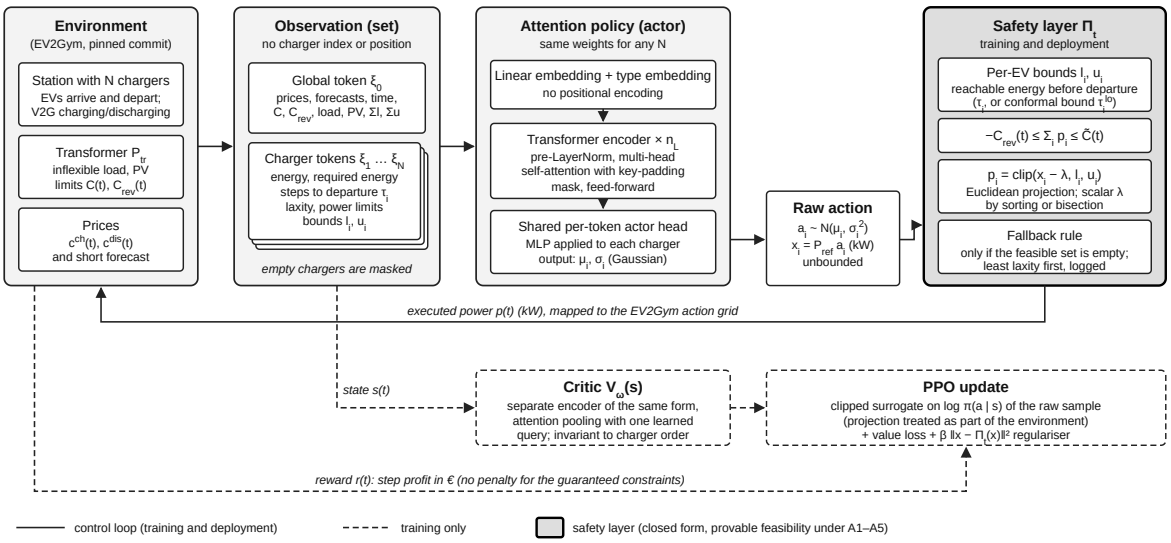


Figure 1: Architecture and control loop of SA-Attn. Solid arrows form the control loop used in training and deployment; dashed arrows are used in training only. The safety layer (shaded) is a closed-form projection with a provable feasibility guarantee under Assumptions A1–A5.

key is never masked, so every softmax is well defined. Each charger output h_i then enters a shared actor head, a two-layer perceptron that returns the mean $\mu_{\theta,i}$ and the standard deviation $\sigma_{\theta,i}$ of a Gaussian raw action. The policy is therefore the product over occupied chargers,

$$\pi_\theta(a \mid s) = \prod_{i \in \mathcal{N}(t)} \mathcal{N}\big(a_i; \mu_{\theta,i}(s), \sigma^2_{\theta,i}(s)\big). \tag{11}$$

The critic has a separate encoder of the same form, pools all unmasked token outputs with one learned attention query [49] and maps the result to the value $V_\omega(s)$.

The construction is permutation-equivariant, masked tokens do not influence the outputs of unmasked tokens, and the number of parameters does not depend on N (Lemma S1 in the supplementary material). With these settings the actor has 610,370 parameters and the critic 676,481, whatever the number of chargers. A station with N chargers of which n are occupied therefore gives the same per-EV outputs as a station with only those n chargers, apart from the global ratios. That property is the structural basis for applying weights trained at $N \in \{10, 20\}$ to $N = 200$, but it does not guarantee performance there, because attention over 200 tokens averages over more keys than seen in training.

4.2 Per-EV bounds from reachable energy

The safety layer needs, for each connected EV, the least power that keeps its departure energy requirement reachable, and we drop the index i where no confusion arises. Full-power charging for one step maps the energy E to

$$\varphi(E) = E + \Delta t\, \eta^{\mathrm{c}}\, \bar{P}(E). \tag{12}$$

Assumption (A1, monotone maximum-charging map)**.** The map φ is continuous and non-decreasing on $[E^{\min}, E^{\mathrm{cap}}]$.

Under A1, starting a step with more energy never leaves less energy after full-power charging, and A1 holds whenever the taper lowers the power by at most $1/(\eta^{\mathrm{c}}\Delta t)$ kW per extra kWh. Let $\Psi_k = \varphi^k$ denote the k-fold composition, with Ψ_0 the identity, and define the reachable-energy operator recursively by

$$R(0, \cdot) = 0, \qquad R(k, E) = \Delta t\, \eta^{\mathrm{c}} \bar{P}(E) + R\big(k-1, \varphi(E)\big), \tag{13}$$

so that $\Psi_k(E) = E + R(k, E)$ is the energy after k full-power steps. The backward-reachable thresholds, the least energies from which the requirement is still reachable, are

$$B(0) = E^{\mathrm{req}}, \qquad B(k) = \min\big\{E \in [E^{\min}, E^{\mathrm{cap}}] :\ \varphi(E) \geq B(k-1)\big\}, \quad k \geq 1. \tag{14}$$

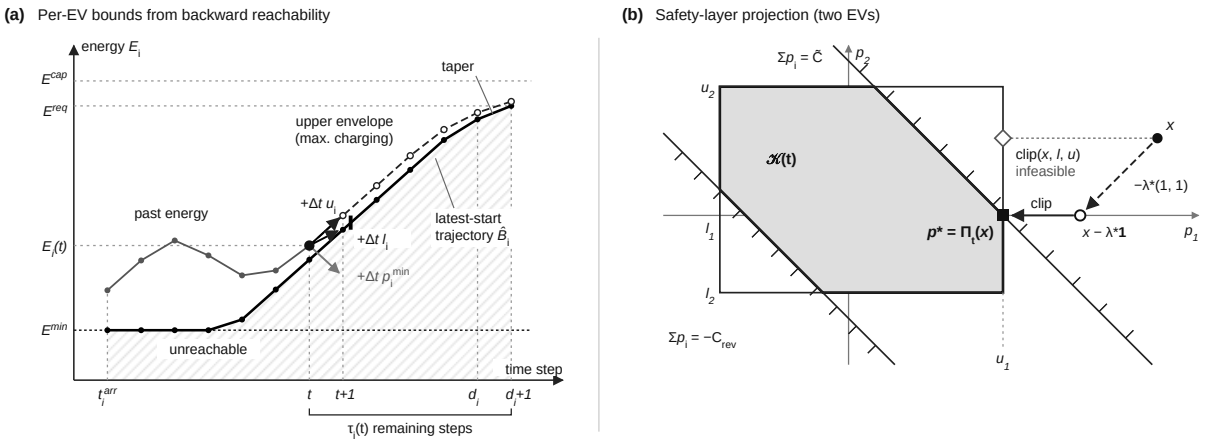


Figure 2: Concepts of the safety layer. (a) Energy of one EV during its stay. The latest-start trajectory $\hat{B}_i$ marks the least energy from which full-power charging still reaches E^{req} by departure; states in the hatched region below it cannot meet the requirement. From the current energy $E_i(t)$, the steps $\Delta t\, u_i$, $\Delta t\, l_i$ and $\Delta t\, p_i^{\min}$ show maximum charging, the lowest admissible power and maximum discharging. The dashed upper envelope is maximum charging with taper, and $\tau_i(t)$ counts the remaining steps. (b) Projection for two EVs. The feasible set $\mathcal{K}(t)$ (shaded) is the box $[l, u]$ cut by the slab $-C_{\mathrm{rev}} \le p_1 + p_2 \le \tilde{C}$. Clipping the raw action x to the box alone gives an infeasible point; the projection shifts x by $-\lambda^*(1, 1)$ and then clips, which gives $p^* = \Pi_t(x)$ on the line $p_1 + p_2 = \tilde{C}$.

Under A1, Ψ_k is non-decreasing, and no admissible power sequence from $E(t)$ ends above $\Psi_k(E(t))$ after k steps (Lemma S2). An EV is therefore individually feasible at step t, in the sense that some admissible control meets Equation 5, exactly when $E(t) \ge B(\tau(t))$. Without A1 this characterisation fails, because charging less now could leave more reachable energy later. For EV2Gym's exponential taper, Lemma S5 proves A1 for every step length whenever the EV's charging power does not exceed the charger rating, a condition met by all eight EV models of the simulator.

For a connected EV with energy $E = E_i(t)$ and $\tau = \tau_i(t)$ remaining actions, the bounds of the safety layer are

$$u_i(t) = \bar{P}(E), \qquad l_i(t) = \begin{cases} \max\Big\{p^{\min}(E),\ g^{-1}\Big(\dfrac{\hat{B}(\tau-1) - E}{\Delta t}\Big)\Big\}, & E \ge \hat{B}(\tau), \\ \bar{P}(E), & E < \hat{B}(\tau), \end{cases} \tag{15}$$

where $\hat{B}$ is the stored threshold table described below, and the first case applies to individually feasible EVs. There, l_i is the smallest admissible power after which $\tau - 1$ full-power steps still reach E^{req}, whereas an individually infeasible EV is charged at maximum power. Without taper, the thresholds have the closed form $B(k) = \max\{E^{\min}, E^{\mathrm{req}} - k\,\eta^{\mathrm{c}} \bar{P}^{\mathrm{nom}} \Delta t\}$, and in the EV2Gym instantiation the lower bound becomes

$$l_i = \max\Big\{ -\min\Big(P^{\mathrm{dis}}, \frac{E - E^{\min}}{\Delta t}\Big),\ \frac{E^{\mathrm{req}} - E}{\Delta t} - (\tau - 1)\bar{P}^{\mathrm{nom}}\Big\}. \tag{16}$$

A negative second term allows V2G discharging down to the energy from which $\tau - 1$ full-power steps still reach E^{req}. As seen in Figure 2(a), the bound l_i keeps the energy on or above the latest-start trajectory and therefore permits discharging only while slack remains.

With a taper, the table $\hat{B}_i(k)$ for $k = 0, \ldots, \tau_i(t_i^{\mathrm{arr}})$ is computed once at arrival by bisection on φ_i, so the per-step cost of Equation 15 is $O(1)$ per EV. Each stored table is certified in the sense that

$$\begin{aligned} \hat{B}(0) \ge E^{\mathrm{req}}, \qquad & \varphi\big(\hat{B}(k)\big) \ge \hat{B}(k-1), \\ \hat{B}(k) \le \max\{\hat{B}(k-1), E^{\min}\}, \qquad & \hat{B}(k) \in [E^{\min}, E^{\mathrm{cap}}] \qquad (k \ge 1). \end{aligned} \tag{17}$$

Any table with the first two properties dominates the exact thresholds, and the proofs use only these properties. The laxity $\chi_i(t) = \big(E_i(t) - \hat{B}_i(\tau_i(t))\big) / \big(\eta_i^{\mathrm{c}} \bar{P}_i^{\mathrm{nom}} \Delta t\big)$, measured in steps, is non-negative exactly when the EV is individually feasible, and it serves as a token feature and as the priority of the fallback rule.

4.3 Projection

Let $x \in \mathbb{R}^N$ be the raw action in kW, with $x_i = l_i = u_i = 0$ for empty chargers, and let $\tilde{C}(t) = C(t) - n(t)\varepsilon_q$ be the tightened forward limit explained in Section 4.4. The feasible set is then

$$\mathcal{K}(t) = \{p \in \mathbb{R}^N : \; l \le p \le u, \; -C_{\mathrm{rev}}(t) \le \mathbf{1}^\top p \le \tilde{C}(t)\}, \tag{18}$$

and the safety layer is the Euclidean projection

$$\Pi_t(x) = \arg\min_{p \in \mathcal{K}(t)} \tfrac{1}{2}\|p - x\|_2^2. \tag{19}$$

Proposition 1 (Projection). *(i) The set $\mathcal{K}(t)$ is non-empty if and only if $l \le u$, $\sum_i l_i \le \tilde{C}$, $\sum_i u_i \ge -C_{\mathrm{rev}}$ and $-C_{\mathrm{rev}} \le \tilde{C}$.*

(ii) Let $\mathcal{S}(\lambda) = \sum_i \mathrm{clip}(x_i - \lambda, l_i, u_i)$. If $\mathcal{K}(t) \neq \emptyset$, the unique minimiser of Equation 19 is $p_i^* = \mathrm{clip}(x_i - \lambda^*, l_i, u_i)$. The shift λ^* is zero if $\mathcal{S}(0) \in [-C_{\mathrm{rev}}, \tilde{C}]$. Otherwise, it is a root of $\mathcal{S}(\lambda) = \tilde{C}$ when $\mathcal{S}(0) > \tilde{C}$ and a root of $\mathcal{S}(\lambda) = -C_{\mathrm{rev}}$ when $\mathcal{S}(0) < -C_{\mathrm{rev}}$.

(iii) Let $\mathcal{F} = \{i : l_i < x_i - \lambda^* < u_i\}$ be the free set and $D = \mathrm{diag}(\mathbb{1}[i \in \mathcal{F}])$. Where $\mathcal{F}$ and the active constraint are locally constant, the Jacobian is $\partial p^*/\partial x = D - (D\mathbf{1})(\mathbf{1}^\top D)/(\mathbf{1}^\top D\mathbf{1})$ if the aggregate constraint is active and $\mathcal{F} \neq \emptyset$. It is D if the aggregate constraint is inactive.

The proof in the supplementary material verifies the Karush–Kuhn–Tucker (KKT) conditions of the strictly convex problem, with λ^* as the multiplier of the aggregate constraint. As seen in Figure 2(b), the projection shifts all free coordinates by the same amount and clips them until the aggregate limit holds. Computing λ^* needs one sort of the $2N$ breakpoints $\{x_i - u_i\} \cup \{x_i - l_i\}$ and one pass over them, so the layer costs $O(N \log N)$ per step. Floating-point rounding needs care when $\sum_i l_i$ equals $\tilde{C}$ up to rounding, so the implementation checks the aggregate limit after each projection. A failing row is recomputed by a bisection that keeps the feasible end of its bracket. Algorithm 1 lists one complete step of the safety layer.

The algebra of Proposition 1 is not new: the problem in Equation 19 is the continuous quadratic knapsack problem, solved by breakpoint algorithms [24] and by a semismooth Newton method [25]. Its clipping form $\mathrm{clip}(x_i - \lambda, l_i, u_i)$ with a common level found by one-dimensional search is also the rate rule of sLLF [19]. Our use differs in its input, a learned per-charger action of a size-agnostic policy, and in the lower bound l_i from backward reachability with taper and V2G sign range. Moreover, the aggregate constraint is two-sided, the layer acts during training, and it carries the multi-step argument of Section 4.4.

4.4 Feasibility invariance

Besides A1, Theorem 1 needs four assumptions on the data, the simulator, the arrivals and the aggregate limits.

Assumption (A2, known data). The departure step d_i and the requirement E_i^{req} are known at arrival.

Assumption (A3, one-step model). For every connected EV and every commanded power p_i within the limits of Equation 3, the realised energy satisfies $E_i(t) + \Delta t\, g_i(p_i) \le E_i(t+1) \le E_i^{\mathrm{cap}}$, the realised power satisfies $p_i \le \tilde{p}_i \le p_i + \varepsilon_q$, and empty chargers draw zero power.

Assumption (A4, individual feasibility at arrival). Every EV is individually feasible when it arrives, that is, $E_i^{\mathrm{arr}} \ge \hat{B}_i\big(\tau_i(t_i^{\mathrm{arr}})\big)$ holds for its stored threshold table.

Assumption (A5, aggregate feasibility). At every step of the closed loop, the bounds and limits satisfy $\sum_i l_i(t) \le \tilde{C}(t)$, $\sum_i u_i(t) \ge -C_{\mathrm{rev}}(t)$ and $\tilde{C}(t) \ge -C_{\mathrm{rev}}(t)$.

In the exact model $\varepsilon_q = 0$, but EV2Gym rounds the normalised action to five decimals and ignores discharge currents below 0.01 A. We therefore round commands upwards on the action grid and use a margin of $\varepsilon_q = 0.008$ kW that covers both effects, at a cost of at most 1.6 kW of forward capacity at $N = 200$. Lemma S6 in the supplementary material lists the simulator settings under which A3 holds in EV2Gym, and the cap of Equation 4 enforces A4 by construction.

Algorithm 1 One step of the safety layer

Require: raw action $x \in \mathbb{R}^N$; for each connected EV: $E_i(t)$, $\tau_i(t)$, certified table $\hat{B}_i$; limits $C(t)$, $C_{\mathrm{rev}}(t)$

for each occupied charger i **do**
  $u_i \leftarrow \bar{P}_i(E_i)$; $p_i^{\min} \leftarrow$ Equation 3
  if $E_i \geq \hat{B}_i(\tau_i)$ **then** $l_i \leftarrow \max\{p_i^{\min}, g_i^{-1}((\hat{B}_i(\tau_i - 1) - E_i)/\Delta t)\}$
  else $l_i \leftarrow u_i$
  end if
  $\chi_i \leftarrow (E_i - \hat{B}_i(\tau_i))/(\eta_i^{\mathrm{c}} \bar{P}_i^{\mathrm{nom}} \Delta t)$
end for
$\tilde{C} \leftarrow C - n(t)\varepsilon_q$
if $\sum_i l_i \leq \tilde{C}$ **and** $\sum_i u_i \geq -C_{\mathrm{rev}}$ **and** $\tilde{C} \geq -C_{\mathrm{rev}}$ **then** ▷ A5 holds
  $p^0 \leftarrow \mathrm{clip}(x, l, u)$
  if $\mathbf{1}^\top p^0 \in [-C_{\mathrm{rev}}, \tilde{C}]$ **then** $p \leftarrow p^0$
  else
    $\vartheta \leftarrow \tilde{C}$ if $\mathbf{1}^\top p^0 > \tilde{C}$, else $-C_{\mathrm{rev}}$
    sort breakpoints $\{x_i - u_i\} \cup \{x_i - l_i\}$; find λ^* with $\mathcal{S}(\lambda^*) = \vartheta$ by one pass
    $p \leftarrow \mathrm{clip}(x - \lambda^*, l, u)$; if $\mathbf{1}^\top p \notin [-C_{\mathrm{rev}}, \tilde{C}]$, recompute by safe bisection
  end if
else
  $p \leftarrow$ fallback rule of Section 4.5; log the case
end if
send p to the simulator, rounded upwards on its action grid
return p

Lemma 1 (One-step invariance)**.** *Assume A1 and A3, and let $\hat{B}$ satisfy Equation 17. Let EV i be connected at step t with $\tau = \tau_i(t)$ remaining actions and energy $E = E_i(t) \geq \hat{B}(\tau)$. Then $p^{\min}(E) \leq l_i(t) \leq u_i(t)$, and every commanded power $p \in [l_i(t), u_i(t)]$ leads to $E_i(t+1) \geq \hat{B}(\tau - 1)$.*

Theorem 1 (Feasibility invariance)**.** *At every step t, let the commanded powers be $p(t) = \Pi_t(x(t))$ for an arbitrary finite raw action $x(t)$, during training or deployment. Under A1–A5, the following statements hold for every step t and every EV i.*

(a) The realised powers satisfy the transformer limit $-C_{\mathrm{rev}}(t) \leq \sum_i \tilde{p}_i(t) \leq C(t)$ at every step, whatever the raw action.

(b) The commanded powers satisfy the physical limits of Equation 3 for every connected EV at every step.

(c) Every EV meets its departure energy requirement $E_i(d_i + 1) \geq E_i^{\mathrm{req}}$ when it leaves the station.

Proof sketch. The proof is an induction over t on the invariant $\mathrm{I}(t)$: $E_i(t) \geq \hat{B}_i(\tau_i(t))$ for all $i \in \mathcal{N}(t)$, and new arrivals satisfy their part of the invariant by A4. If $\mathrm{I}(t)$ holds, Lemma 1 gives valid bounds, and A5 with Proposition 1(i) makes $\mathcal{K}(t)$ non-empty. The projection then returns a point of $\mathcal{K}(t)$, from which claim (b) follows directly and claim (a) follows through the margin of A3. Lemma 1 carries every connected EV to $\mathrm{I}(t+1)$, and at $\tau = 1$ it yields $E_i(d_i + 1) \geq \hat{B}_i(0) \geq E_i^{\mathrm{req}}$, so claim (c) holds. The supplementary material gives the full proof. □

The theorem holds for any raw action, so it covers the random exploration of training as well as deployment. A1 enters the proof exactly once, in the implication $E \geq \hat{B}(\tau) \Rightarrow \varphi(E) \geq \varphi(\hat{B}(\tau))$, and a certified table does not compensate for a violation of A1.

Corollary 1 (Localisation)**.** *Assume A1–A4. Every EV that receives $p_i(t) \geq l_i(t)$ at every step of its stay meets its departure energy requirement, whatever happens at other steps; in particular, every EV whose stay contains no infeasible step meets it. The physical limits hold at every step, and the transformer limit holds at every step except in cases (i), (ii) and (iv) of the fallback rule in Section 4.5.*

Assumption A5 is a closed-loop condition, because $l_i(t)$ depends on the energies that the policy has produced, and per-EV bounds cannot guarantee it. For example, two EVs that each need 10 kWh within

two 1-h steps under a 10 kW limit can share the capacity equally. A policy that idles in the first step, however, leaves $\sum_i l_i = 20$ kW in the second step. Policy-independent sufficient conditions for A5 exist, but they ask the transformer to serve all constrained EVs at full power at once (supplementary material). We therefore state the guarantee conditionally on A5, localise it per EV, and count how often A5 fails in the experiments.

4.5 Fallback rule and admission control

When the feasibility test of Proposition 1(i) fails, the fallback rule replaces the projection, and since $l \le u$ holds by construction, four exclusive cases remain.

(i) Contradictory limits, $\tilde{C} < -C_{\mathrm{rev}}$: the procedure of case (iii) is applied with priority on the forward limit.

(ii) Excess PV export, $\sum_i u_i < -C_{\mathrm{rev}}$: every occupied charger receives $p_i = u_i$.

(iii) Excess required power, $\sum_i l_i > \tilde{C}$: every EV starts at $p_i^{\min}$, and the budget $b = \tilde{C} - \sum_i p_i^{\min}$ is distributed in increasing order of laxity, each EV receiving up to $l_i - p_i^{\min}$.

(iv) Negative budget within case (iii): the output stays at the maximal discharge $p = p^{\min}$.

Case (iii) is the least-laxity-first fallback rule, and individually infeasible EVs have negative laxity and therefore come first. It keeps the transformer limit, and every feasible EV that receives l_i stays feasible (Lemma S3). In our scenarios the PV peak stays below the rating, so case (ii) cannot occur, and since EV2Gym clips the inflexible load at the rating, case (iv) needs $C(t) < n(t)\varepsilon_q$. The resulting departure shortfall of an episode is bounded a priori by

$$\Delta t \sum_{t \in \mathcal{T}_{\mathrm{iii}}} \Big(\sum_i l_i(t) - \tilde{C}(t) \Big) + \Delta t \sum_{t \in \mathcal{T}_{\mathrm{iv}}} \sum_i \big(l_i(t) - p_i^{\min}(t) \big), \tag{20}$$

where $\mathcal{T}_{\mathrm{iii}}$ and $\mathcal{T}_{\mathrm{iv}}$ are the steps of cases (iii) and (iv) (Lemma S4).

Admission control lowers the frequency of case (iii) by checking each arrival in advance. When an EV arrives, a small feasibility LP checks whether all connected EVs can still reach their targets under the forecast transformer limits. The LP starts from the current energies and replaces the taper by a linear inner approximation. Inside this LP only, the target of an EV that is already connected is its admitted requirement, lowered where necessary to the maximum energy that the LP's taper model can deliver. Without this adjustment, full-battery targets would make the LP infeasible whenever such an EV is connected, because the linear taper model never reaches full charge in finitely many steps. If the LP is infeasible, the request of the new EV is lowered to the largest feasible value, but requests of EVs already admitted are never reduced. The reduced request passes through the same certified routine as the cap of Equation 4, so A4 still holds. Admission control is a heuristic, because it relies on forecasts and ignores future arrivals, and it does not enter Theorem 1.

4.6 Conformal lower bound on the departure step

Assumption A2 requires the departure step at arrival, and we relax it for d_i with split conformal prediction (CP) [60,61,66,67] in a one-sided, quantile-regression form [68]. A gradient-boosting regressor $\hat{q}$ estimates a lower quantile of d_i from features $\mathbf{w}_i$, namely the arrival step, the weekday, the arrival energy, the requested energy and the EV model. The regressor is fitted on separate simulated sessions, and calibration draws n_{cal} sessions from independently seeded episodes of the test configuration, one per episode and chosen uniformly at random. With the scores $S_j = \hat{q}(\mathbf{w}_j) - d_j$, the conformal quantile $\hat{Q}_{1-\alpha}$ is the k-th smallest score with $k = \lceil (n_{\mathrm{cal}} + 1)(1 - \alpha) \rceil$, and the bound for a test EV is

$$\hat{d}_i^{\mathrm{lo}} = \max \big\{ t_i^{\mathrm{arr}},\ \lceil \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha} \rceil \big\}, \qquad \tau_i^{\mathrm{lo}}(t) = \max\{1,\ \hat{d}_i^{\mathrm{lo}} - t + 1\}. \tag{21}$$

The safety layer, the cap and the token features then use τ_i^{lo} in place of τ_i, and a departure is observed when it occurs.

Proposition 2 (Conformal departure guarantee)**.** *Let the calibration sessions and the test session be exchangeable.*

(i) $\mathbb{P}\big(d_i \geq \hat{d}_i^{\mathrm{lo}}\big) \geq 1-\alpha$.

(ii) Under A1, A3 and A4 with τ^{lo} *in place of* τ, $\mathbb{P}\big(E_i(d_i{+}1) \geq E_i^{\mathrm{req}}\big) \geq 1-\alpha-\mathbb{P}\big(\exists\, t \leq d_i :\ p_i(t) < l_i(t)\big)$. *If A5 holds almost surely at every step of the stay, the probability is at least* $1-\alpha$.

The probability is over the calibration data and a uniformly chosen EV of a test episode, so it equals the expected within-episode fraction of covered EVs. It is neither a per-EV guarantee nor a guarantee conditional on $\mathbf{w}_i$. Calibration is repeated for every test configuration, including every N, because the law of a uniformly chosen session depends on the configuration. A conservative bound has a price in larger lower bounds, more capped requests and lower profit, and the experiments report this price for $\alpha \in \{0.01, 0.05, 0.1, 0.2\}$.

4.7 Training algorithm

The policy is trained with proximal policy optimization (PPO) [69] and generalised advantage estimation [70], and the safety layer, the fallback rule and the actuation map are treated as part of the environment. PPO therefore uses the likelihood ratio of the raw Gaussian sample a, and for this MDP the score-function gradient is unbiased for the return of the projected actions. For stochastic policies, projection inside the environment and inside the policy are equivalent in this sense [46]. Our choice avoids the bias of scoring a projected action under the unprojected density [44] and the zero-gradient problem of back-propagating through an active projection [45].

Two masks reduce variance without biasing the score-function gradient: tokens with $u_i - l_i \leq 10^{-9}$ kW cannot influence any executed power and are dropped from the log-probability and the entropy. Transitions of fallback steps do not depend on the action, so they are masked out of the actor loss but kept for the critic, and the masked loss is normalised by the fixed minibatch size. Many raw actions map to the same projected point, so the policy mean can drift far outside $\mathcal{K}(t)$ without changing the return. A regulariser with weight $\beta = 0.01$ counters this drift,

$$\mathcal{L}_{\mathrm{reg}}(\theta) = \beta\, \mathbb{E}_s\Big[\frac{1}{n(t)} \sum_{i \in \mathcal{N}(t)} \Big(\frac{\bar{x}_i - \Pi_t(\bar{x})_i}{P^{\mathrm{ref}}}\Big)^2\Big], \qquad \bar{x} = P^{\mathrm{ref}} \mu_\theta(s). \tag{22}$$

By the Moreau identity, the gradient of $\frac{1}{2}\mathrm{dist}^2(x, \mathcal{K})$ is $x - \Pi_t(x)$, so $\Pi_t(\bar{x})$ can be evaluated under stop-gradient and no Jacobian of the projection is needed. Distance terms of this kind are an established remedy against drift [15, 46]. Each rollout runs sixteen environments, eight with $N = 10$ chargers and eight with $N = 20$, and token tensors of different N are padded with masked entries at update time. Algorithm S1 in the supplementary material gives the full training loop.

5 Experimental Setup

5.1 Simulator and scenarios

All experiments use EV2Gym [6] at commit `55ef368c` without changes to its source. A wrapper computes the observations, applies the safety layer, the fallback rule and the actuation map, logs all metrics and never reads future arrivals. Each station has N chargers with one port each behind a single transformer, and EV specifications are drawn from the eight V2G-capable EV models of the simulator with one constant efficiency per model. The desired energy is the full battery, prices are the Dutch day-ahead prices shipped with EV2Gym, and the discharge price equals the charging price. Table S1 in the supplementary material lists all scenario parameters, and Section S2 documents the modelling details.

Every exogenous quantity scales linearly with N. The transformer rating is $P_{\mathrm{tr}} = \rho_{\mathrm{tr}} N$ with a tight level of $\rho_{\mathrm{tr}} = 4$ kW per charger, the ratio of EV2Gym's 25-charger V2G configuration, and a moderate level of 7 kW. Inflexible load and PV are scaled to a mean peak of $0.4 P_{\mathrm{tr}}$, so the PV peak stays below the rating. Forecasts in the observation carry Gaussian noise with a standard deviation of 10% of the value.

Three arrival profiles of EV2Gym are used, namely workplace, public and private (residential). Their start hours and spawn multipliers are chosen so that overnight sessions fit into the 28-h horizon and the mean charger occupancy is about 0.21 for all profiles. The policy does not observe the profile, so the profile forms a further zero-shot dimension.

Training mixes $N \in \{10, 20\}$, the three profiles and the two ratings within every batch. The test grid covers $N \in \{5, 10, 20, 50, 100, 200\}$, both ratings and all profiles, i.e. 36 configurations with 100 fixed test episodes each for $N \leq 50$ and 50 for $N \geq 100$. EV2Gym's original 25-charger configuration adds 100 episodes for comparability with earlier EV2Gym studies, so each evaluated policy runs 3,100 test episodes shared across methods. Two sensitivity studies at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW cover all three profiles. The first evaluates the main policies with a desired state of charge of 0.8, and the second trains a separate policy with $\Delta t = 5$ min and $\gamma = 0.99^{1/3}$.

5.2 Compared methods

The learned baselines isolate the two parts of SA-Attn: Attention+penalty uses the same attention policy without the safety layer and learns from a penalty reward. A multilayer perceptron (MLP) with two hidden layers of 256 units over the concatenated tokens, zero-padded to 200 slots, replaces the attention policy in MLP+safety layer and in MLP+penalty. Without the safety layer, a raw action passes only the per-EV physical clip $\mathrm{clip}(x_i, p_i^{\min}, u_i)$, and that clip contains neither the lower bound l_i nor the transformer coupling. The penalty reward subtracts 10 € per kWh of departure shortfall and per kWh of transformer overload energy, more than eleven times the highest hourly price in the data, so no violation can pay.

Four heuristics complete the comparison: as fast as possible (AFAP), as late as possible (ALAP), round robin and LLF, each in a raw form and with its output passed through the safety layer. The economic MPC of EV2Gym [30] runs with true prices and arrivals inside its 10-step horizon, an information advantage that is flagged in the results. Its commercial solver licence limits it to $N \leq 25$, so larger stations use a related but not identical HiGHS port, reported under its own name. EV-GNN [11] runs from its public code at commit `01aa1b7a` with its own TD3 training script, reward and published hyperparameters. Its script accepts one scenario per run, so one model per arrival profile is trained at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW with five seeds and evaluated zero-shot on all sizes and ratings of that profile. GNN-DT [12] is not included, because we found no public code for it. The perfect-information LP of Section 3.6 provides the reference profit $J^{\mathrm{LP,ceil}}$.

Ablations of SA-Attn remove admission control, set $\beta = 0$, train with a curriculum or use a smaller network. The conformal variant SA-Attn-CP(α) is evaluated at all four levels of α.

5.3 Metrics

Per episode we record the profit in € and the optimality gap of Equation 9. Service is measured by the departure shortfall in kWh and the share of EVs that miss E_i^{req} by more than 0.01 kWh, and grid safety by the transformer violation energy in kWh. For the safety layer we count the infeasible steps per fallback case and record the deficit bound of Equation 20 and the number of capped and admission-reduced requests. The cost of the guarantee is the profit of Attention+penalty minus the profit of SA-Attn on the same episodes, reported together with the violations that the penalty method incurs. For the conformal variant, coverage is the episode-averaged fraction of EVs with $d_i \geq \hat{d}_i^{\mathrm{lo}}$.

5.4 Training, tuning and statistics

All learned methods of our own code receive the same tuning budget of 18 configurations over learning rate, minibatch size and entropy coefficient, each trained for 0.5×10^6 steps on the tight 10-charger station. The selected configuration is retrained with five seeds for 3×10^6 steps, and variants such as SA-Attn-CP inherit the selection of their parent method. Table S2 in the supplementary material lists all hyperparameters and the tuning grid. In total, the main experiments, ablations and penalty-weight sensitivities comprise 51 training runs of our own PPO code, plus the tuning runs and the optional EV-GNN and 5-min runs.

Table 2: Main comparison at the training station sizes $N \in \{10, 20\}$ (three arrival profiles and both tightness levels pooled; placeholder until the analysis scripts have run).

Method	Profit [€]	Gap [%]	IQM gap [%]	Unserved [%]	Violation [kWh]	Infeasible
SA-Attn (ours)	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Attention+penalty	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
MLP+safety layer	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
MLP+penalty	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
EV-GNN / +safety layer	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Heuristics / +safety layer	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Economic MPC / HiGHS port	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]

Aggregate scores over seeds and episodes are reported as the mean and the interquartile mean with stratified-bootstrap 95% confidence intervals [71], following the reproducibility guidance of [72]. Paired comparisons on common episodes use the Wilcoxon signed-rank test [73] on per-episode profit and shortfall, with Holm correction across N. The implementation uses PyTorch [74], a single-file PPO in the style of CleanRL [75] and the Adam optimiser [76], with the safety layer in double precision. Code, configurations and analysis scripts will be released with the paper, pinned to the EV2Gym and EV-GNN commits above [TODO: repository link]. [TODO: Add the hardware (CPU, GPU, memory, operating system, Python and PyTorch versions) used for the reported runs].

6 Results

[TODO: Skeleton: fill every red result placeholder from the analysis outputs, use the past tense and write the interpretation yourself].

6.1 Main comparison

Table 2 compares all methods on the test configurations that match the training sizes, $N \in \{10, 20\}$, pooled over profiles and ratings. For each method the table reports the profit, the optimality gap, the share of unserved EVs, the transformer violation energy and the number of infeasible steps. At these sizes, the optimality gap of SA-Attn is [RESULT: mean gap and 95% confidence interval at $N \in \{10, 20\}$]. Relative to Attention+penalty, its profit differs by [RESULT: difference in € per episode and Wilcoxon p-value], and the penalty method leaves [RESULT: unserved EVs and violation energy of Attention+penalty]. Among the heuristics with the safety layer, [RESULT: best heuristic and its gap], and the economic MPC reaches [RESULT: gap of the MPC, with its information advantage noted]. Methods whose profit exceeds $J^{\mathrm{LP,ceil}}$ through transformer violations are [RESULT: list of such methods, or "none"].

6.2 Zero-shot scaling

Figure 3 plots the profit per charger and the optimality gap against the number of chargers, with weights trained at $N \in \{10, 20\}$ applied without retraining up to $N = 200$. At $N = 50$, 100 and 200, the gap of SA-Attn is [RESULT: gap with 95% confidence interval at each size]. From the training sizes to $N = 200$, the gap changes by [RESULT: change in percentage points], against [RESULT: change for MLP+safety layer and for EV-GNN]. Across arrival profiles and transformer ratings, [RESULT: differences between the three profiles and the two ratings], and Table S3 in the supplementary material lists all values. [TODO: Interpret the scaling trend in your own words and relate it to the size independence of Section 4.1].

6.3 Constraint satisfaction

Figure 4 shows the energy not served at departure and the transformer violation energy per method and station size. On feasible steps, the methods with the safety layer record a transformer violation energy of [RESULT: value from the logs; Theorem 1 predicts zero]. The penalty-based methods leave [RESULT: shortfall and violation energy per N]. Infeasible steps occur in [RESULT: fraction of steps per N, profile and rating, split into fallback cases], and the measured shortfall amounts to [RESULT: shortfall as a

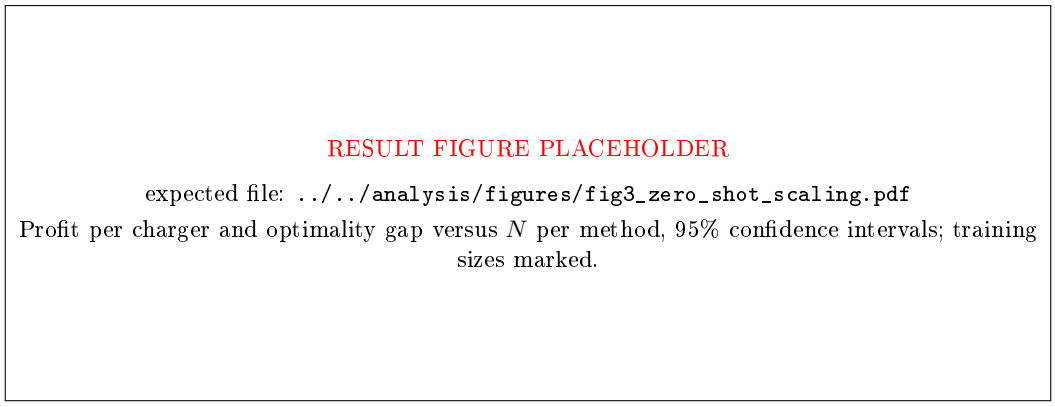


Figure 3: Zero-shot scaling of the learned methods and of the heuristics with the safety layer for N from 5 to 200.

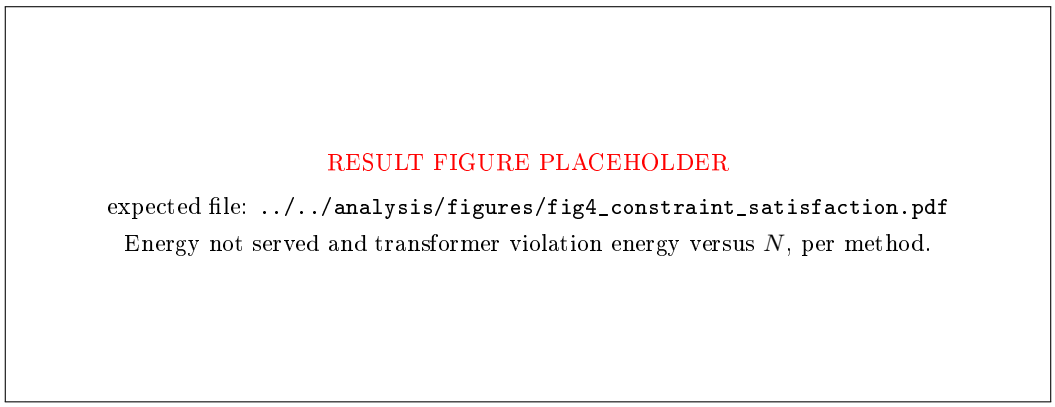


Figure 4: Constraint satisfaction across station sizes.

fraction of the deficit bound of Equation 20]. Requests are capped by more than 0.01 kWh for [RESULT: number of EVs], and admission control reduces [RESULT: number of requests and kWh], as listed in Table S4.

6.4 Cost of the guarantee

Figure 5 plots the cost of the guarantee against N, together with the violations of the penalty-trained policy. A positive cost means that the penalty-trained policy earned more profit than SA-Attn on the same episodes. Per EV, the cost of the guarantee is [RESULT: cost in € per EV and in percent of $|J^{\mathrm{LP,ceil}}|$ at each N]. For that profit, the penalty-trained policy incurs [RESULT: shortfall and violation energy], and with penalty weights of 1 and 100 €/kWh the cost becomes [RESULT: values of the sensitivity runs]. [TODO: Interpret the trade-off in your own words].

6.5 Further results

The supplementary material reports the remaining analyses, and their main values are summarised here. In the ablation study (Table S6), removing admission control changes the share of infeasible steps by [RESULT: value], and setting $\beta = 0$ changes the gap by [RESULT: value]. For the conformal variant (Table S7), the episode-averaged coverage is [RESULT: coverage for each α], at a profit loss of [RESULT: loss relative to known departure steps]. At $N = 200$, one control step takes [RESULT: mean inference time of policy and safety layer] (Table S8), and the sensitivity studies give [RESULT: gap at desired state of charge 0.8 and at $\Delta t = 5$ min]. An example episode at $N = 20$ in Figure S6 shows [RESULT: how aggregate power follows the price and the limits].

RESULT FIGURE PLACEHOLDER

expected file: `../../analysis/figures/fig6_cost_of_guarantee.pdf`

Cost of the guarantee (profit of the penalty-based policy minus profit with the safety layer) versus N, with the violations of the penalty-based policy.

Figure 5: Cost of the guarantee across station sizes.

7 Discussion

7.1 Interpretation

[TODO: Paragraph 1: state whether weights trained at 10 and 20 chargers transfer to 200 chargers (Figure 3, Table 2) and compare the gaps]. The main finding on zero-shot transfer is [RESULT: one-sentence summary].

[TODO: Paragraph 2: relate the cost of the guarantee (Figure 5) to the violations of the penalty-trained policy, i.e. lost profit versus profit from violations]. The main finding on the cost of the guarantee is [RESULT: one-sentence summary].

[TODO: Paragraph 3: discuss how often A5 failed and the shortfall left by the fallback rule (Figure 4), relating it to the closed-loop nature of A5]. The main finding on infeasible steps is [RESULT: one-sentence summary].

[TODO: Paragraph 4: compare with Table 1 only on the literature configuration, never across different simulators or scenarios].

7.2 Practical implications for charge point operators

This subsection is to be written by the authors after the experiments, covering the following points.

- [TODO: What a CPO gains when the guarantees hold for any raw action, e.g. for certification or grid-operator agreements.]
- [TODO: The cost of the guarantee in euros per EV and per year for a station of given size.]
- [TODO: Use of the conformal variant where drivers do not declare departure times; acceptable coverage levels.]
- [TODO: Inference time of Table S8 relative to the 15-min control step.]

7.3 Limitations

The guarantee of Theorem 1 rests on known departure times and requested energies (A2). Our conformal extension relaxes only the departure step, and it gives a marginal coverage statement rather than a per-EV guarantee. Requested energies are still assumed to be declared at arrival, and strategic misreporting of deadlines [58] lies outside the model.

Aggregate feasibility (A5) can fail in closed loop, as the two-EV example of Section 4.4 shows. On such steps the fallback rule keeps the transformer limit and spreads the deficit in order of laxity, so the departure guarantee holds only for the EVs covered by Corollary 1. Lemma S4 bounds the remaining

shortfall but does not remove it. A layer that certifies joint feasibility over the future would need time-coupled constraints, and its projection would lose the single-sum form.

The model covers one station behind one transformer, and it ignores voltage limits, line flows and other distribution-network constraints. Several transformers or a network model would add further coupled sum constraints to $\mathcal{K}(t)$, a case that Proposition 1 does not cover.

Our experiments use EV2Gym's lossless accounting, in which the billed power equals the battery-side energy change. V2G arbitrage therefore carries no round-trip loss, so profits from discharging may be optimistic. Efficiencies are constant per EV model instead of following the current-dependent tables of the simulator. The theory covers lossy, piecewise-linear energy maps, but the experiments do not exercise them.

Assumption A1 requires a monotone maximum-charging map. Lemma S5 proves A1 for the exponential taper of EV2Gym, provided the charging power of the EV does not exceed the charger rating. A taper whose power drops faster than $1/(\eta^c \Delta t)$ kW per kWh would violate A1, and certified thresholds would not repair the guarantee. Exact delivery of sub-maximal commands (A3) also depends on the default taper multiplier of the simulator.

All results come from simulation with synthetic arrivals drawn from EV2Gym's statistics and with Dutch day-ahead prices. Some late evening sessions of the public and private profiles do not fit into the episode horizon and are therefore not generated. Forecast noise is synthetic, and real metering delays are not modelled.

Our EV-GNN baseline is trained per arrival profile at a single size with its own algorithm, so that comparison is not fully like-for-like. The same holds for the MPC beyond 25 chargers, a related but not identical port. In addition, the perfect-information LP relaxes the charging taper to the nominal power, so the reported optimality gap overestimates the true gap. Part of any gap is also the price of not knowing the future, because the LP ignores the information limits of an online controller. Finally, the novelty claims depend on the full texts of the closest works listed as open risks in the related-work review. [TODO: Remove or adapt the last sentence once the novelty risks are closed].

8 Conclusion

This paper combined a size-agnostic attention policy with a closed-form safety layer for V2G charging control at a CPO station. Each charger is a token, and one set of weights serves any number of chargers. The safety layer projects raw actions onto per-EV bounds derived from a reachable-energy operator and onto the two-sided transformer limit, using the known form of a common shift with clipping in $O(N \log N)$ time. Under stated assumptions, Theorem 1 guarantees the transformer limit and every departure energy requirement at every step for any raw action, so the guarantee holds during training as well as deployment. A least-laxity-first fallback rule, admission control and a deficit bound handle the steps on which aggregate feasibility fails. For uncertain departures, a split-conformal bound on the departure step replaces the known-departure assumption at a stated coverage level.

In the zero-shot evaluation, [RESULT: one or two sentences on the optimality gap up to $N = 200$]. Regarding constraint satisfaction, [RESULT: one sentence on violations, shortfall and the frequency of infeasible steps]. The cost of the guarantee was [RESULT: one sentence on its size relative to the profit and to the violations avoided].

Future work can extend the feasible set to several transformers and to linearised voltage constraints, at the price of coupled sum constraints beyond the single-sum form used here. A second direction is a layer that certifies joint feasibility over the future, for example through nested deadline constraints, so that the closed-loop assumption on aggregate feasibility is no longer needed. Finally, a conformal bound on the requested energy and tests with recorded charging sessions would narrow the gap between the model and practice.

Article Information Form

Authors' Contributions
[TODO: State the contribution of each author, for example "Author 1: conceptualisation, methodology, software, writing; Author 2: supervision, review and editing"; the renewed DergiPark system may also ask for a contributor-role statement].

Acknowledgments
[TODO: Acknowledge contributors who do not meet the authorship criteria, or write "Not applicable"]. The authors used an artificial intelligence assistant during the preparation of this work, as detailed in the Artificial Intelligence Statement below.

Conflict of Interest
The authors declare that there is no conflict of interest regarding the publication of this paper. [TODO: Confirm this declaration].

Support/Supporting Organizations
[TODO: Name every source of financial support with grant numbers, or state that no financial support was received].

Ethical Approval
[TODO: Confirm the following wording]. This study is a simulation study that involves no human participants, no human data collected by the authors and no animals, so ethics committee approval was not required. The authors declare that they complied with research and publication ethics.

Availability
The code, the scenario configurations, the test replays and the analysis scripts will be released in a public repository. [TODO: Repository link, licence and archived version]. The experiments use the public EV2Gym code at commit `55ef368c` and, for one baseline, the public EV-GNN code at commit `01aa1b7a`.

Artificial Intelligence Statement
The authors used an artificial intelligence (AI) assistant, Claude (Anthropic), model claude-opus-5-5, during the preparation of this work for three purposes. First, it supported literature searching and the screening of candidate references, and the authors then checked every reference against the publisher record via its DOI. Second, it assisted with code development for the simulation experiments. Third, it suggested preliminary wording for parts of the manuscript and improved the language. All AI-suggested text was treated as a preliminary draft and was thoroughly reviewed, rewritten and verified by the authors. No AI tool was used to generate, report or interpret the research results, and the authors take full responsibility for the content of this article. [TODO: Confirm the final wording, the tool name and version, and that every statement is accurate; repeat the disclosure in the Methods or Acknowledgments as the journal's AI policy requires].

Plagiarism Statement
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